Lecture 32: Unsupervised Machine Learning (Part-2) — Dimensionality Reduction & Anomaly Detection (Hinglish)

Swagat hai **Lecture 32: Unsupervised Machine Learning (Part-2)** mein. Yeh lecture modern unsupervised machine learning ke do bohot zaroori topics ko cover karta hai:

1. **Dimensionality Reduction**: Principal Component Analysis (PCA), Curse of Dimensionality, Hughes Phenomenon aur mathematical eigen-decomposition.
2. **Anomaly & Outlier Detection**: Density-based, tree-based aur local density algorithms: DBSCAN, Isolation Forest aur Local Outlier Factor (LOF).

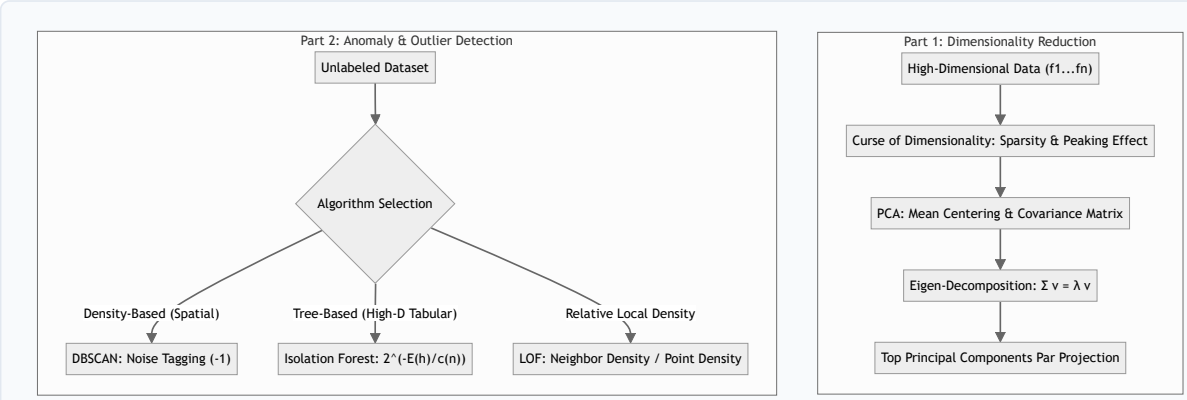
Complete Curriculum & Submodule Navigation

Submodule	Topic & Title	Type	Code Files	Description
`Notes_32.01`	**Dimensionality Reduction & Curse of Dimensionality**	Theory	—	Data sparsity ($O(k^d)$), Euclidean distance ka be-asar hona, computation cost, Hughes effect aur feature selection vs extraction.
`Notes_32.02`	**What is PCA - Intuition & Principles**	Theory	—	Netflix user engagement example, correlation ki zaroorat, spread $\equiv$ variance, mean centering, PC1 aur perpendicular PC2.
`Notes_32.03`	**PCA Step-by-Step Mathematical Algorithm**	Theory	—	Z-score scaling, covariance matrix Σ, Pythagoras theorem derivation ($\min d_1 \iff \max d_2$), $\Sigma\mathbf{v} = \lambda\mathbf{v}$ aur projection matrix W.
`Notes_32.04`	**PCA Implementation on Iris Dataset**	Practical	`pca.ipynb`	Scikit-Learn `PCA(n_components=2)`, explained variance ratio (72.96% & 22.85%), eigenvector weights aur 2D scatter plot visualization.
`Notes_32.05`	**What is Anomaly Detection -**	Theory	—	Asli antar: Outliers (extreme par valid) vs Anomalies (suspicious violations); real-world

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	Outlier vs Anomaly			domains: banking fraud, factory QC, medical tests, botnet.
Notes_32.06	**DBSCAN for Anomaly Detection**	Theory	—	Density-based noise separation, core vs border vs noise (`label = -1`), aur K-Means ki forced assignment se comparison.
Notes_32.07	**DBSCAN Implementation for Anomaly Detection**	Practical	clustering_DBSCAN.ipynb	Non-linear `make_moons` data par DBSCAN, hyperparameter tuning rules ($\text{min_samples} \approx 2d = 5, \epsilon = 0.18$) aur noise visualization.
Notes_32.08	**Isolation Forest for Anomaly Detection**	Theory	—	Isolation trees (iTrees), random cuts, path length $h(x)$, anomaly scoring formula $s(x, n) = 2^{-E(h(x))/c(n)}$ aur BST average depth.
Notes_32.09	**Isolation Forest on Clinical Thyroid Dataset**	Practical	isolation_forest.ipynb thyroid_dataset.csv	6,916 patients ke 21 attributes par clinical anomaly detection, `contamination=0.036` ke sath 249 hypothyroid pathology outliers detect karna.
Notes_32.10	**Local Outlier Factor (LOF) for Anomaly Detection**	Theory	—	Multi-density clusters, local vs global outliers ka antar, k-distance, reachability distance, local reachability density (LRD) aur LOF ratio.
Notes_32.11	**Local Outlier Factor Implementation**	Practical	local_outlier_factor.ipynb thyroid_dataset.csv	21-feature clinical data par LOF implementation, demographic local anomalies aur 2D PCA visual scatter plot.
Practice_Problems	**Practice Case Studies & Solutions**	Problems & Solutions	Solutions_HINGLISH.md	3 syllabus-aligned real-world case studies: E-Commerce PCA analytics, semiconductor wafer defect screening

Submodule	Topic & Title	Type	Code Files	Description
				aur clinical thyroid diagnostics.
`Lecture_Code`	**Master Official Code Directory**	Code Directory	Saare Notebooks & CSV	Ma'am ke dwara diye gaye official original notebooks aur datasets ka master index.

Overall Architecture Flow



Important Rules Followed

- **Pure Theory Modules Mein Koi Faltu Code Notebook Nahi**: Modules `32.01`, `32.02`, `32.03`, `32.05`, `32.06`, `32.08`, `32.10` mein sirf concepts, mathematical formulas aur diagrams hain.
- **Official Code Fidelity**: Code modules (`32.04`, `32.07`, `32.09`, `32.11`) mein tested clean Jupyter notebooks hain jo Ma'am ke padhaye code ke 100% anuroop hain.
- **MathJax Column 0 Alignment**: Saare display math formulas `$$` column 0 par unindented hain taaki Markdown aur PDF dono mein bina error render hon.
- **Bilingual Notes & Vector PDFs**: Har sub-module mein English aur Hinglish dono versions hain aur sabhi ke vector-quality PDFs generate kiye ja rahe hain.